

The enduring contradictions of new software development approaches: a response to 'Persistent Problems and Practices in ISD'

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In 'Persistent Problems and Practices in Information Systems Development', Kautz *et al.* (2007) offer a thoughtful and original perspective on the nature of high-speed software development practices. They offer controversy over the degree to which the set of software development practices, variously called agile methodology, internet speed methodology and short cycle time development, are different from more traditional or plan-based development approaches. We heartily welcome such published research and conference debates that reflect the various views on this issue (Glass 2003; Merisalo-Rantanen *et al.*, 2005).

Kautz *et al.* are provocative. They lead us to wonder how history might regard software development agility. Will agile methods become a milestone in software development? Will these methods mark a bifurcation of practice where agile methods will have offered a revolutionary alternative to plan-based approaches? Or, will, instead, agile methods someday be remembered as a bogus bait-and-switch? Have unwitting developers, desperate to overcome incredibly competitive market conditions, actually adopted traditional plan-based systems development practices in a deluded belief that these veteran practices were somehow novel and revolutionary?

Kautz *et al.* (2007) build insights from their argumentation and empirical research that call for more extensive frameworks for considering these practices. They provide us with an example of such an alternative framing of the problems and practices. Through this different analytical framework, they illustrate how accentuated evolution is less than a revolution. This analysis finds that the software development problems are basically the same as always and that the 'new' practices are also basically the same as always. Such a result would indeed contradict our findings (Baskerville *et al.*, 2003).

This is a new genre of software engineering process. Though many of the characteristics of the development processes may be observed in traditional software development, the intensity with which they apply and together distinguish Internet software development. In that sense, Internet software development uses a process that is different from more traditional software development processes, but surprisingly similar within its new genre (Ramesh *et al.*, 2002, p. 170).

We suspect there are at least three basic reasons for the contradiction. First, the phenomena being circumscribed for comparison may not be exactly the same. In hindsight, our use of the term 'Internet speed' and 'Internet software development' reflected the language among practitioners at the time. Others have used the term 'agile development' and indeed we have also used 'short cycle time' and 'high speed' development, terms borrowed from engineering, in a search for the best label for the approaches we studied (Baskerville & Pries-Heje, 2004; Baskerville *et al.*, 2006). From our perspective, Web development which is the focus of Kautz *et al.* (2007) and the agile development approaches we studied are not equivalent. Agile development does not necessarily imply the use of Web interfaces, and is successfully used for other kinds of applications, such as developing process control applications, developing middleware software and other non-web applications. Similarly, the development of Web-based software does not necessarily imply agile development, and Web applications have been developed successfully using decidedly plan-based approaches, such as will frequently be the case in banking and finance.

Further, significant distinguishing characteristics of Web projects are not necessarily found in agile projects. Kautz *et al.* (2007) identify the following as important/distinguishing features of Web applications: 'look and feel, user interface, marketing capability, multimedia content and content production'. In contrast, the applicability of high-speed practices is typically evaluated based on the following dimensions: number of people, criticality and project priorities (Cockburn, 2000). For example, these practices have been used successfully by small teams, in non-mission critical projects which emphasize speed of development as a top priority. While these practices are likely to work well in many web development projects, they may not be appropriate, for example, in mission critical Web projects. Second, the application of different analytical frameworks, even to the same underlying phenomena, can lead to contradictory results. For example, they analyse the practices using dimensions of coping behaviour regarding diversity, knowledge and structure. These dimensions are indeed excellent intellectual tools for drawing forwards the similarity of software practices across time. In our studies, we have used dimensions of coping behaviour regarding time, cost and scope. These dimensions, long established as the three major constraints of software development (Humphrey, 1988), have proven useful in highlighting the distinguishing characteristics of more recent practices. Kautz *et al.* (2007) provide a framework that highlights the similarities and, as a contrast to ours, that highlights the differences; we regard their work as an insightful extension of research such as ours, that has explored both the similarities and the differences among these classes of methods (Baskerville *et al.*, 2003).

We have provided strong evidence within our analytical frames that agile development is different in fundamental ways. We agree the problems are essentially the same, but we find the solutions are indeed different because of the intense focus on time. In particular, the combination of practices and the intensity with which they are applied is different. In Baskerville *et al.* (2003), we link principles and practices, and show how some old practices have been discarded in agile approaches, how some old practices have been adopted into the new framework, and how some practices are entirely new. Older practices certainly persist in agile approaches and Kautz *et al.*'s (2007) research helps us to understand this persistence better.

Aside from these practical aspects of the contrasting development approaches, however, there are even deeper strategic and economic differences that result from the availability and applications of these approaches. For example, software project managers can align their portfolio of projects with development strategies in new ways, representing the equivalent software development alternatives to the production model (traditional) versus a job shop model (agile) (Slaughter *et al.*, 2006).

Third, we do not think that the definitions or analytical frames applied by the researcher are the ultimate determinant of whether the problems and practices are the same or different. Our claims rely strongly on the beliefs held by the software developers we studied, and their quite certain assertions that they regarded their practices as different.

We develop a lot based on out-of-the-box applications. It has changed the way we develop systems. We probably spend more time adapting and for example making them more open that we did before.

Competition is driving the speed. We are always looking to keep up with and ahead of the competition.

We basically took a lot of current processes . . . and transferred them to the Internet . . . and found that it didn't work.

It is *their* belief that their practices were different from traditional or plan-based development practices. Across the three phases of this longitudinal study, our respondents included many professionals who had moved from traditional development organizations (or departments within their organizations) to join the agile development groups. Whether the distinction between plan-based and agile is an objective reality, or a socially constructed reality, the effect on software practices is the same nonetheless.

This fundamental underlying belief held by the subjects is reflected as an assumption in our findings. One motive for our research has been our interest in discovering if software developers had simultaneously begun adopting a rather similar set of behaviours across many different organizations, and why they were doing this. They certainly did adopt such a set, and we set about explaining how and why.

While we have succumbed to the temptation to seek ways to resolve the contradictions between the findings and analyses of our study, and those of our esteemed colleagues reported above, it is inevitable that these contradictions will endure both in opinion and in reality. We concur with Robert Glass's evaluation of this debate:

There have always been important differences between software projects. Diversity is a key part of what makes the software world go 'round' . . . Information systems have always been developed differently from scientific systems, and critical projects have been treated differently from more mundane ones. Why should we be surprised that the same condition holds for Web and non-Web projects? There are important differences between the two, . . . but those degrees of difference have always been a vital and even healthy part of what the software world is all about (Glass, 2003, p. 23).

Ultimately we agree with Kautz *et al.*'s (2007) conclusions on two very important points. One, of course, is the need for further research in this area. As the dot-com era recedes into history, such research as Kautz *et al.*'s will help shape our understanding of just how important (or unimportant) the agile movement will prove. The second point is that observed practices may be different from good practices, a point that they offer as insight to improve many empirical studies of software development, including ours.

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